

MATH230: Tutorial Two [Solutions]

Propositional Logic: Natural Deductions with Negation

Key ideas

- Write natural deduction proofs involving $\neg$ negation.
- Write natural deductions using Ex Falso i.e. $\perp$ elimination.

Relevant Topic: Propositional Logic

Relevant reading: L $\exists\forall$ N Chapters 3,4 and Simon section 1.1

Hand in exercises: 1a, 1b, 1c, 1i, 1k

Due Friday @ 5pm to the submission box on Learn.

Discussion Questions

1. Show $A \vdash \neg\neg A$.

2. Show $(A \vee B) \wedge (A \vee C) \vdash A \vee (B \wedge C)$.

Hints

Read these hints and suggestions before reading the solutions below. It is important that you try to solve these problems first, before reading a solution. All hints from earlier labs still apply to this lab.

In this second lab we now use propositions of the form $\neg\alpha$. However, this requires no new rules to deal with the $\neg$ connective as we define $\neg\alpha := \alpha \rightarrow \perp$. Therefore, we need only the implication introduction and elimination rules to deal with negations.

This introduction of $\neg$ and $\perp$ can lead to inconsistent contexts i.e. $\Sigma \vdash \perp$ is possible for some Σ . Different logics deal with the introduction of $\perp$ in different ways. In this tutorial you will be asked to make use of the following $\perp$ -elimination rule called *ex falso quodlibet*.

$$\frac{\begin{array}{c} \Sigma \\ \vdots \\ \perp \\ A \end{array}}{A} \text{XF}$$

$\perp$ Elimination (Ex Falso)

More commonly called the *principle of explosion*, this rule of inference states that if a contradiction can be derived from Σ , then in fact any proposition at all follows.

Hints and Tips

Always unpack $\neg\alpha$ for $\alpha \rightarrow \perp$ to help remember the rules that apply in eliminating and introducing $\neg$ negations.

Tutorial Exercises

1. **Minimal Logic.** Provide natural deduction proofs of the following sequents. These deductions require only the use of minimal logic; the introduction and elimination rules for $\wedge$ conjunction, $\vee$ disjunction, $\rightarrow$ implication, and the definition of $\neg$ negation as an implication.

(a) $\neg A \vdash (C \rightarrow A) \rightarrow \neg C$

Solution:

$$\frac{\frac{\overline{C}^2 \quad \overline{C \rightarrow A}^1}{A} \text{MP} \quad \neg A}{\frac{\perp}{\neg C} \rightarrow I, 2} \text{MP} \rightarrow I, 1$$

(b) $A \wedge \neg B \vdash \neg(A \rightarrow B)$

Solution:

$$\frac{\frac{A \wedge \neg B}{A} \wedge E_L \quad \frac{\overline{A \rightarrow B}^1}{B} \text{MP} \quad \frac{A \wedge \neg B}{\neg B} \wedge E_R}{\frac{\perp}{\neg(A \rightarrow B)} \rightarrow I, 1} \text{MP}$$

(c) $A \rightarrow C, B \rightarrow D, \neg C \vee \neg D \vdash \neg A \vee \neg B$

Solution:

$$\frac{\frac{\overline{\neg C}^2 \quad \overline{A}^1 \quad A \rightarrow C}{C} \text{MP} \quad \frac{\overline{\neg D}^3 \quad \overline{B}^4 \quad B \rightarrow D}{D} \text{MP}}{\frac{\frac{\perp}{\neg A} \rightarrow I, 1 \quad \frac{\perp}{\neg B} \rightarrow I, 4}{\neg A \vee \neg B} \vee I_L \quad \frac{\frac{\perp}{\neg A} \rightarrow I, 1 \quad \frac{\perp}{\neg B} \rightarrow I, 4}{\neg A \vee \neg B} \vee I_R} \rightarrow I, 2 \quad \rightarrow I, 3 \quad \vee E$$

(d) $A, \neg A \vdash \neg B$

Solution:

$$\frac{A \quad \neg A}{\perp} \text{MP} \rightarrow I$$

(e) $A \rightarrow B, A \rightarrow \neg B \vdash \neg A$

Solution:

$$\frac{\frac{\overline{A}^1 \quad A \rightarrow B}{B} \text{MP} \quad \frac{\overline{A}^1 \quad A \rightarrow \neg B}{\neg B} \text{MP}}{\frac{\perp}{\neg A} \rightarrow I, 1} \text{MP}$$

(f) $A \rightarrow \neg B \vdash B \rightarrow \neg A$

Solution:

$$\frac{\frac{\overline{A}^2 \quad A \rightarrow \neg B}{\neg B} \text{ MP} \quad \overline{B}^1}{\frac{\perp}{\neg A} \rightarrow I, 2} \text{ MP} \\ \frac{\perp}{B \rightarrow \neg A} \rightarrow I, 1$$

(g) $\neg(A \wedge B) \vdash A \rightarrow \neg B$

Solution:

$$\frac{\frac{\overline{A}^1 \quad \overline{B}^2}{A \wedge B} \wedge I \quad \neg(A \wedge B)}{\frac{\perp}{\neg B} \rightarrow I, 2} \text{ MP} \\ \frac{\perp}{A \rightarrow \neg B} \rightarrow I, 1$$

(h) $A \vdash \neg\neg A$

Solution:

$$\frac{A \quad \neg\overline{A}^1}{\perp} \text{ MP} \\ \frac{\perp}{\neg\neg A} \rightarrow I, 1$$

(i) $\neg\neg\neg A \vdash \neg A$

Solution:

$$\frac{\frac{\overline{A}^1}{\neg\neg A} \text{ THM} \quad \neg\neg\neg A}{\perp} \text{ MP} \\ \frac{\perp}{\neg A} \rightarrow I, 1$$

(j) $\neg A \vee \neg B \vdash \neg(A \wedge B)$

Solution:

$$\frac{\frac{\overline{A \wedge B}^1}{A} \wedge E_L \quad \neg\overline{A}^2 \quad \frac{\overline{A \wedge B}^1}{B} \wedge E_R \quad \neg\overline{B}^3}{\frac{\perp}{\neg A \rightarrow \perp} \rightarrow I, 2 \quad \frac{\perp}{\neg B \rightarrow \perp} \rightarrow I, 3} \text{ MP} \\ \frac{\perp}{\neg(A \wedge B)} \rightarrow I, 1 \quad \vee E$$

(k) $\neg A \wedge \neg B \vdash \neg(A \vee B)$

Solution:

$$\frac{\frac{\neg A \wedge \neg B}{\neg A} \wedge E_L \quad \overline{A}^2 \quad \frac{\neg A \wedge \neg B}{\neg B} \wedge E_R \quad \overline{B}^3}{\frac{\perp}{A \rightarrow \perp} \rightarrow I, 2 \quad \frac{\perp}{B \rightarrow \perp} \rightarrow I, 3} \text{ MP} \\ \frac{\perp}{\neg(A \vee B)} \rightarrow I, 1 \quad \vee E$$

(l) $\neg(A \vee B) \vdash \neg A \wedge \neg B$

Solution:

$$\frac{\frac{\frac{\overline{A}^1}{A \vee B} \vee I_L \quad \neg(A \vee B)}{\perp} \rightarrow I, 1 \quad \text{MP} \quad \frac{\frac{\frac{\overline{B}^2}{A \vee B} \vee I_R \quad \neg(A \vee B)}{\perp} \rightarrow I, 2 \quad \text{MP}}{\neg A \wedge \neg B} \wedge I$$

(m) $A \rightarrow \neg B \vdash \neg(A \wedge B)$

Solution:

$$\frac{\frac{\frac{\overline{A \wedge B}^1}{A} \wedge E_L \quad A \rightarrow \neg B}{\neg B} \text{MP} \quad \frac{\frac{\overline{A \wedge B}^1}{B} \wedge E_R}{\perp} \text{MP}}{\neg(A \wedge B) \rightarrow I, 1}$$

(n) $\vdash \neg\neg(A \vee \neg A)$

Solution:

$$\frac{\frac{\frac{\overline{A}^2}{A \vee \neg A} \vee I_L \quad \overline{\neg(A \vee \neg A)}^1}{\perp} \text{MP} \quad \frac{\frac{\frac{\perp}{\neg A} \rightarrow I, 2}{A \vee \neg A} \vee I_R \quad \overline{\neg(A \vee \neg A)}^1}{\neg\neg(A \vee \neg A) \rightarrow I, 1} \text{MP}$$

2. **Intuitionistic derivations.** Provide natural deduction proofs of the following. You do not need to use the *classical* $\perp$ rule for these questions, but may find that the *intuitionistic* $\perp$ rule is necessary.

(a) $A, \neg A \vdash B$

Solution:

$$\frac{A \quad \neg A}{\perp} \text{MP} \\ \frac{\perp}{B} \text{XF}$$

(b) $\neg A \vdash A \rightarrow B$

Solution:

$$\frac{\overline{A}^1 \quad \neg A}{\perp} \text{MP} \\ \frac{\perp}{B} \text{XF} \\ \frac{A \rightarrow B}{A \rightarrow B} \rightarrow I, 1$$

(c) $\neg A \vee B \vdash A \rightarrow B$

Solution:

$$\frac{\overline{A}^1 \quad \overline{\neg A}^2}{\perp} \text{MP} \\ \frac{\perp}{B} \text{XF} \\ \frac{\neg A \vee B \quad \frac{\neg A \rightarrow B}{\neg A \rightarrow B} \rightarrow I, 2 \quad B \rightarrow B}{B} \vee E \\ \frac{B}{A \rightarrow B} \rightarrow I, 1$$

(d) $A \vee B, \neg A \vdash B$

Solution:

$$\frac{\overline{A}^1 \quad \neg A}{\perp} \text{MP} \\ \frac{\perp}{B} \text{XF} \\ \frac{A \vee B \quad \frac{A \rightarrow B}{A \rightarrow B} \rightarrow I, 1 \quad B \rightarrow B}{B} \vee E$$

(e) $\vdash \neg(B \rightarrow A) \rightarrow (A \rightarrow B)$

Solution:

$$\frac{\overline{A}^2}{B \rightarrow A} \rightarrow I \quad \frac{\overline{\neg(B \rightarrow A)}^1}{\perp} \text{MP} \\ \frac{\perp}{B} \text{XF} \\ \frac{A \rightarrow B}{A \rightarrow B} \rightarrow I, 2 \\ \frac{A \rightarrow B}{\neg(B \rightarrow A) \rightarrow (A \rightarrow B)} \rightarrow I, 1$$

(f) $A \rightarrow B, A \rightarrow \neg B \vdash A \rightarrow C$

Solution:

$$\frac{\overline{A}^1 \quad A \rightarrow B}{B} \text{MP} \quad \frac{\overline{A}^1 \quad A \rightarrow \neg B}{\neg B} \text{MP} \\ \frac{\perp}{C} \text{XF} \\ \frac{C}{A \rightarrow C} \rightarrow I, 1$$

(g) $A \vee B, \neg A, \neg B \vdash C$

Solution:

$$\begin{array}{c}
 \frac{\neg A \quad \overline{A}^1}{\perp} \text{MP} \quad \frac{\neg B \quad \overline{B}^1}{\perp} \text{MP} \\
 \frac{\perp}{C} \text{XF} \quad \frac{\perp}{C} \text{XF} \\
 \frac{A \vee B \quad \frac{A \rightarrow C}{\rightarrow I, 1} \quad \frac{B \rightarrow C}{\rightarrow I, 2}}{C} \vee E
 \end{array}$$